Comprehensive analysis of AQI in Delhi using Power BI and Power Query

Objective:

To conduct a comprehensive analysis of Air Quality Index (AQI) in Delhi using Power BI and Power Query to identify key pollutant drivers of poor air quality.

Create dashboard including KPIs, AQI Category distribution, Pollutant comparison, Hourly trend and Weekly pattern

Task 1 : Data Preparation Using Power Query

- Changed the column to correct date data type.
- Added time column and day name column.
- Replaced the zero value with mean of the column without zero.

Task 2 : Calculate AQl

PM2.5 AQI formula :

```
AQI_PM2.5 =
SWITCH(TRUE(),
[pm2.5] <=30, [pm2.5]*50/30,
[pm2.5] <=60, 50 + ([pm2.5]-30)*(50/30),
[pm2.5] <=90, 100 + ([pm2.5]-60)*(100/30),
[pm2.5] <=120, 200 + ([pm2.5]-90)*(100/30),
[pm2.5] <=250, 300 + ([pm2.5]-120)*(100/130),
500
)
```

PM10 AQI formula :

```
AQI_PM10 =
SWITCH(TRUE(),
[pm10] <=50, [pm10],
[pm10] <=100, 50 + ([pm10]-50),
[pm10] <=250, 100 + ([pm10]-100),
[pm10] <=350, 200 + ([pm10]-250),
[pm10] <=430, 300 + ([pm10]-350),
500
)
```

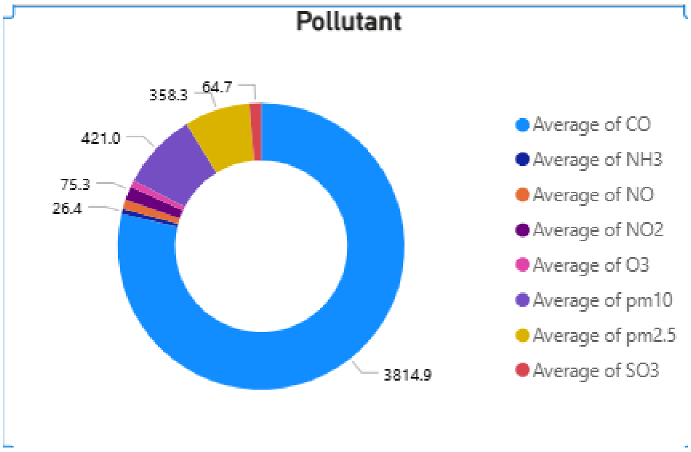
AQI : formula to take highest pollutant AQI

```
AQI = MAXX({[AQI_PM2.5],[AQI_PM10]},[Value])
```

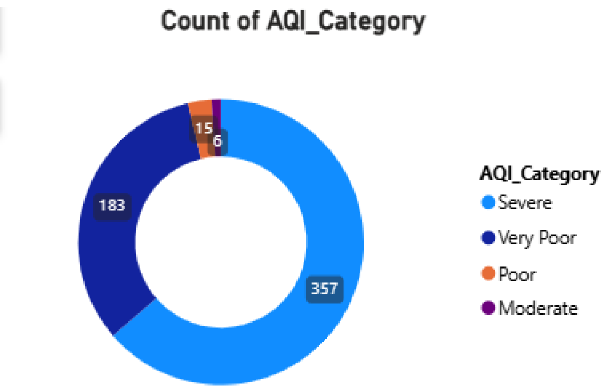
AQI Category Column:

```
AQI_Category =
SWITCH(TRUE(),
[AQI] <=50,"Good",
[AQI] <=100,"Satisfactory",
[AQI] <=200,"Moderate",
[AQI] <=300,"Poor",
[AQI] <=400,"Very Poor",
"Severe")
```

Task 3 : Create PowerBI dashboard

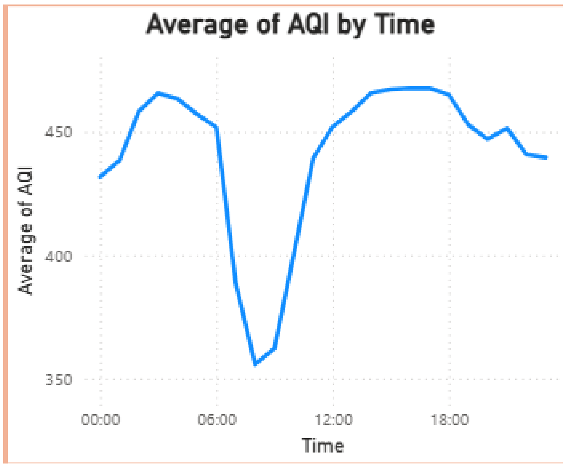


Average concentration by pollutant CO appears dominant.

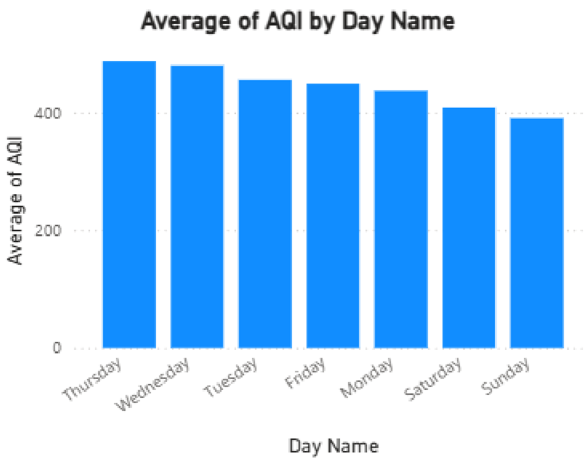


Delhi is in Severe or Very Poor category most of the time.

The majority of recorded days fall under Severe and Very Poor categories, indicating chronic air quality stress.



- AQI high during early morning (4 AM – 6 AM)
- Sharp drop around 7–9 AM
- AQI rises again from 12 PM onwards
- Evening (4 PM – 8 PM) shows peak pollution
- Slight drop late night



AQI is slightly higher during weekdays compared to weekends, with Sunday showing the lowest levels. This suggests vehicular and commercial activity contributes significantly to pollution levels.